

A Differentiable Heston IV Surrogate

End-to-end pipeline, arbitrage-aware training, and
Monte-Carlo validation

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Abstract

We build and evaluate a neural network surrogate for the Heston stochastic volatility model that maps a six-dimensional parameter vector $(\kappa, \theta, \sigma_v, \rho, v_0, r)$ to a 49×14 grid of Black–Scholes implied volatilities. Training data are produced by a Fourier-cosine (COS) pricer using the Albrecher *little trap* characteristic function; the training loss combines a vega-weighted IV-RMSE term with soft arbitrage constraints (calendar, Durrleman butterfly, term-structure slope). An independent Monte-Carlo reference built from Andersen’s quadratic-exponential (QE) scheme with maturity-adaptive step counts anchors the noise floor at 3.44 bps. The final surrogate reaches a validation IV-RMSE of 35.99 bps and a vs-MC IV-RMSE of 36.67 bps on 84 747.00 pricable cells; the 50.00 bps gate holds on 89.58% of cells, missing the 95% target driven by a thin p_{99} tail. Post-calibration inversion from a flat prior recovers v_0, θ, σ_v cleanly while κ, ρ, r reflect the classical Heston identifiability limits rather than a surrogate flaw. A non-obvious LayerNorm-after-Dropout invariance bug, which inflated the eval-mode validation loss by $3.2\times$, is documented in full.

Contents

1	Introduction	2
2	Heston model and Fourier pricing	2
2.1	Risk-neutral dynamics	2
2.2	Characteristic function (Albrecher “little trap”)	3
2.3	COS pricer	3
2.4	Implied volatility inversion	3
3	Data generation	4
3.1	Realistic Scenario-C prior	4
3.2	Strike–maturity grid	4
3.3	Quality mask and pricable region	4
3.4	Dataset composition	5
4	Monte-Carlo reference	5
4.1	Andersen QE scheme	5
4.2	Maturity-adaptive step count	6
5	Surrogate architecture	6
5.1	Stem and residual trunk	6
5.2	Grid-factorised head	6
5.3	Pricable-region buffer	6

6	Training	7
6.1	Vega-weighted IV-RMSE loss	7
6.2	Soft arbitrage constraints	7
6.3	Optimiser, schedule and mixed precision	7
6.4	A non-obvious bug: LayerNorm after Dropout	8
7	Results	8
7.1	Training convergence	8
7.2	Surface examples	9
7.3	Gate vs MC reference	9
7.4	Calibration sanity	10
8	Discussion and future work	11
8.1	What this project establishes	11
8.2	What it does not yet achieve	11
8.3	Reproducibility	12

1 Introduction

The Heston [1] stochastic volatility model remains a workhorse for equity-index derivatives. Calibration (the inverse problem of recovering $(\kappa, \theta, \sigma_v, \rho, v_0)$ from observed IV surfaces) is expensive because each forward evaluation requires semi-analytic Fourier integration across a strike-maturity grid. A neural surrogate that approximates the grid-to-grid map [7, 8, 9, 10, 11]

$$\Phi : (\kappa, \theta, \sigma_v, \rho, v_0, r) \longmapsto \{\sigma^{\text{BS}}(k_i, T_j)\}_{i=1, \dots, N_K; j=1, \dots, N_T}$$

replaces each Fourier call with a single dense-network evaluation and yields analytic gradients $\partial\Phi/\partial\theta$ for free, which in turn makes calibration a straightforward gradient-descent problem.

This report documents the end-to-end pipeline of the *A Deep Calibration* workflow: (i) the COS pricer and characteristic function used to generate training surfaces (Section 2), (ii) the data-generation pipeline, prior, grid, and quality mask (Section 3), (iii) the independent MC reference (Section 4), (iv) the surrogate architecture and loss (Section 5), (v) the training recipe including a subtle LayerNorm-after-Dropout bug uncovered mid-project (Section 6), and (vi) empirical results on held-out data, the MC gate, and calibration sanity (Section 7). We close with limitations (Section 8).

2 Heston model and Fourier pricing

2.1 Risk-neutral dynamics

Under the pricing measure $\mathbb{Q}$, the Heston model specifies

$$dS_t = (r - q) S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \tag{1a}$$

$$dv_t = \kappa(\theta - v_t)dt + \sigma_v \sqrt{v_t} dW_t^{(2)}, \quad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt. \tag{1b}$$

We set $q \equiv 0$ and take r as the flat risk-free rate over the surface horizon. The Feller condition $2\kappa\theta \geq \sigma_v^2$ is *not* enforced in our prior: it is routinely violated in real equity calibrations and our quality mask (Section 3) handles the ensuing short-maturity numerical edge cases explicitly.

2.2 Characteristic function (Albrecher “little trap”)

Let $X_t = \log S_t$ and $\varphi_T(u) = \mathbb{E}[e^{iuX_T} \mid X_0, v_0]$. The *little-trap* formulation of [2] fixes the branch-cut issues of the original Heston paper while preserving the closed form. Define

$$\begin{aligned}\xi &= \kappa - i\rho\sigma_v u, \\ d &= \sqrt{\xi^2 + \sigma_v^2} u(u + i), \\ g &= \frac{\xi - d}{\xi + d}, \\ A(u, T) &= iu(r - q)T, \\ B(u, T) &= \frac{\kappa\theta}{\sigma_v^2} \left[(\xi - d)T - 2 \log \frac{1 - g e^{-dT}}{1 - g} \right], \\ C(u, T) &= \frac{v_0}{\sigma_v^2} \frac{(\xi - d)(1 - e^{-dT})}{1 - g e^{-dT}},\end{aligned}$$

so that

$$\varphi_T(u) = \exp(A + B + C).$$

All computation is carried out in `np.complex128` with no branch-tracking logic required.

2.3 COS pricer

Call and put prices are evaluated by the Fourier-cosine expansion of [3]: on a log-price interval $[a, b]$,

$$C(x_0, T) \approx e^{-rT} \sum'_{n=0}^{N-1} \Re \left[\varphi_T \left(\frac{n\pi}{b-a} \right) e^{-in\pi a/(b-a)} \right] U_n,$$

where the primed sum halves the $n = 0$ term and U_n are the payoff coefficients (calls use the standard formulae; *puts use their own payoff coefficients directly* to avoid catastrophic cancellation against a deep-ITM call-parity reconstruction, which was an unforeseen blocker resolved over the course of the project).

Adaptive truncation. The half-width L that defines $[a, b] = [c_1 + x_\ell - L, c_1 + x_\ell + L]$ is set *per cell* using the Heston cumulants

$$c_1 = (r - q)T - \frac{1}{2} \int_0^T \mathbb{E}[v_s] ds, \quad c_2 = \theta T + (v_0 - \theta) \frac{1 - e^{-\kappa T}}{\kappa},$$

via

$$L = \max \left(0.5, |c_1 + x_\ell| + L_{\text{fac}} \sqrt{c_2} \right),$$

with $L_{\text{fac}} = 10$ in the standard tier and $L_{\text{fac}} = 15$ in the extended tier. Adaptive truncation prevents short- T under-resolution at the wings.

Two-tier dispatch. A cell is priced with $N_{\text{COS}} = 512$; if the resulting price-to-spot ratio falls below 10^{-4} or is non-finite, the extended tier ($N_{\text{COS}} = 2048$) is tried. Cells with relative resolution below 10^{-6} are marked unpricable.

2.4 Implied volatility inversion

Prices are converted to Black–Scholes implied volatilities by a safeguarded Newton–Raphson iteration with a rational warm start. Convergence failures (deep OTM puts, short T , very small prices) propagate into the quality mask rather than being silently clipped.

3 Data generation

3.1 Realistic Scenario-C prior

Training and evaluation samples are drawn from a six-dimensional *realistic* prior designed to cover SPX-like calibrations without excluding stressed regimes:

Parameter	Lower	Upper	Distribution
κ	0.30	10.00	log-uniform
θ	0.01	0.16	log-uniform (long-run $\sigma \in [10\%, 40\%]$)
σ_v	0.10	1.50	uniform
ρ	-0.90	-0.30	truncated normal $\mathcal{N}(-0.6, 0.3^2)$
v_0	0.02	0.20	log-uniform (initial $\sigma \in [14\%, 45\%]$)
r	0.00	0.05	uniform

Samples are drawn with a scrambled Sobol sequence of dimension five (plus an independent uniform draw for r) to achieve quasi-uniform coverage of the unit hypercube. The Feller constraint $2\kappa\theta \geq \sigma_v^2$ is *not* imposed.

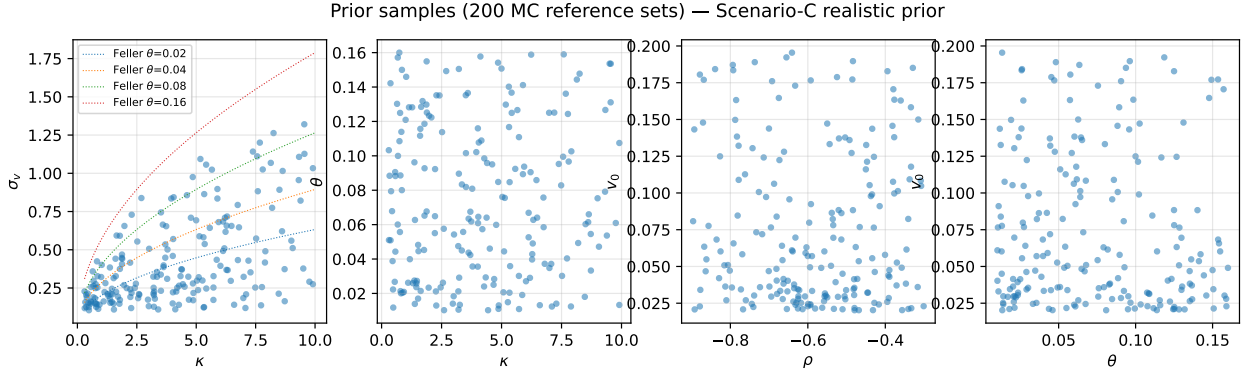


Figure 1: Projections of the $N = 200$ MC-reference parameter sets onto four planes of interest. The left panel shows the Feller envelope $\sigma_v = \sqrt{2\kappa\theta}$ for several values of θ (dotted lines): a substantial fraction of the support sits above the envelope, i.e. violates Feller.

3.2 Strike-maturity grid

Surfaces are represented on a 49×14 grid:

- **Log-moneyness:** 49 equispaced nodes on $[-0.80, +0.40]$ ($\Delta k \approx 0.025$); $k = \log(K/S)$.
- **Maturity (years):** $\{1/52, 2/52, 3/52\} \cup \{1, 2, 3, 4, 6, 9\}/12 \cup \{1.00, 1.25, 1.50, 2.00, 3.00\}$.

This yields $N_K N_T = 686$ cells per surface. The grid is deliberately dense at the wings and at the short end, where numerical edges of both the COS pricer and the MC reference appear.

3.3 Quality mask and pricable region

Each cell is tagged with one of four levels:

- 0 — **cos_standard:** COS priced in the standard tier, IV inversion converged.
- 1 — **cos_extended:** COS priced in the extended tier, IV inversion converged.
- 2 — **cos_small_price:** price at the numerical edge, IV inversion unreliable.
- 3 — **unpricable:** pricer failure or non-finite IV.

Training uses only cells with $\text{qm} \leq 1$ (V2_QM_GOOD_MAX); others carry weight zero. The *pricable region* $\mathcal{P} \subset \{1, \dots, N_K\} \times \{1, \dots, N_T\}$ is the intersection of $\text{qm} \leq 1$ cells across a calibration survey of the prior and is stored as a persistent buffer in the surrogate (Figure 2); it has $424/686 = 61.8\%$ coverage.

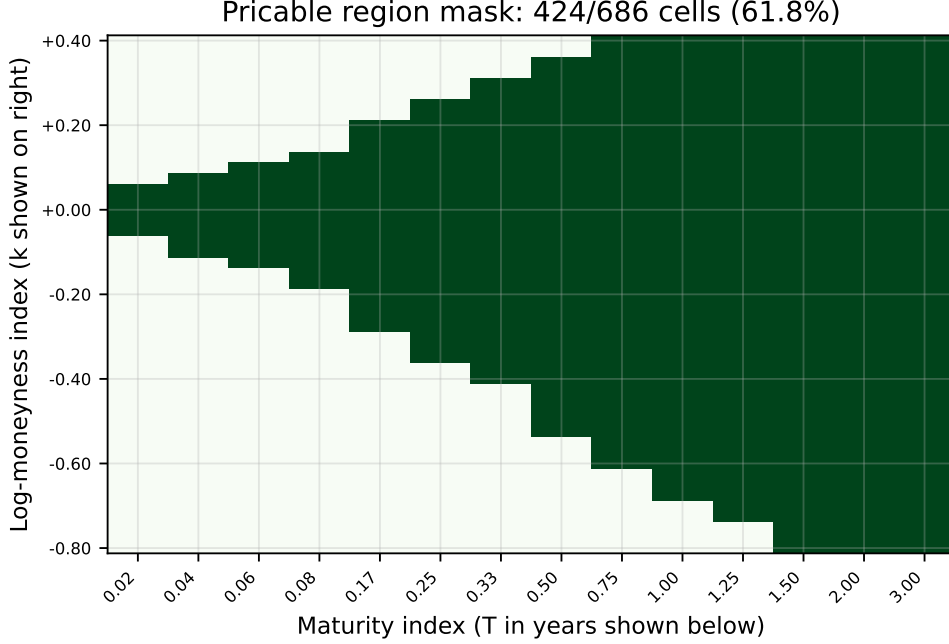


Figure 2: Pricable region $\mathcal{P}$: 424/686 cells (61.8%). Non-pricable cells (white) cluster at short maturities, deep OTM puts (large negative k), and deep ITM calls — exactly where the COS window becomes degenerate.

3.4 Dataset composition

The v2 training HDF5 stores 160 000.00 parameter sets (train split) and 20 000.00 sets (validation split). Fields: `params` ($N, 6$), `iv_surface` ($N, 49, 14$), `quality_mask` ($N, 49, 14$) $\in \{0, 1, 2, 3\}$, `pricable_region` ($49, 14$). All inputs are normalised to $[0, 1]$ by the training dataset loader using the bounds of Section 3.1; a zero-valued dividend channel $q \equiv 0$ is appended for future multi-asset extension.

4 Monte-Carlo reference

An independent MC reference is required because any training-set validation is circular: the training loss and the validation loss both use the pricer as ground truth. The MC reference exercises the underlying SDE (1) via a completely different numerical pathway.

4.1 Andersen QE scheme

We use Andersen’s quadratic-exponential (QE) scheme [4]. For a step from t to $t + \Delta t$, let

$$m = \theta + (v_t - \theta)e^{-\kappa\Delta t}, \quad s^2 = \frac{v_t \sigma_v^2 e^{-\kappa\Delta t} (1 - e^{-\kappa\Delta t})}{\kappa} + \frac{\theta \sigma_v^2 (1 - e^{-\kappa\Delta t})^2}{2\kappa},$$

and $\psi = s^2/m^2$. If $\psi \leq \psi_c = 1.5$ we use the quadratic branch $v_{t+\Delta t} = a(b + Z)^2$ with $Z \sim \mathcal{N}(0, 1)$ and a, b chosen to match m, s^2 ; otherwise we use the exponential branch $v_{t+\Delta t} = F^{-1}(U)$ with $U \sim \mathcal{U}(0, 1)$ and F a mixture of a point mass at 0 and an exponential tail. The log-stock step uses the Andersen K_0^* martingale correction and an antithetic pair for variance reduction.

4.2 Maturity-adaptive step count

The original Phase 2A reference used a fixed $\mathbf{sy} = 252$ steps per year, which under-resolves the short- T / high- κ corner and biased the MC IVs by tens of basis points there. The Phase 3 reference (used throughout this report) instead sets $\mathbf{sy}$ per maturity as

$$\mathbf{sy}(T) = \max\left(\mathbf{sy}_{\min}, \lceil c_{\text{qe}} \cdot T^{-1/2} \rceil\right),$$

so that the short end receives up to 5000.00 steps per year while long maturities remain computationally cheap. With 1 000 000.00 paths per surface and 200.00 parameter sets, the MC IV matches the COS pricer to an IV-RMSE of 3.44 bps ($p_{95} = 6.10$ bps, $p_{99} = 11.30$ bps) on the pricable region. This constitutes the *noise floor* below which no fair gate can be demanded of the surrogate.

5 Surrogate architecture

The surrogate $\Phi_\phi : \mathbb{R}^7 \rightarrow \mathbb{R}^{N_K N_T}$ (six physical inputs plus a zero-valued dividend channel) is implemented in `model.network.BatesSurrogate` with ≈ 6.57 million trainable parameters.

5.1 Stem and residual trunk

The stem is a single linear projection to width $2W$ (with $W = 512$), LayerNorm, and SwiGLU activation; the trunk is $N_B = 6$ pre-norm ResBlocks each of the form

$$x \leftarrow x + \text{SiLU}(\text{Linear}_{W \rightarrow W} \text{SwiGLU}(\text{Linear}_{W \rightarrow 2W}(\text{LayerNorm}(x)))).$$

Crucially the trunk contains *no* dropout: see Section 6.4 for the reason.

5.2 Grid-factorised head

The dominant head exploits the grid structure: given a trunk embedding $h \in \mathbb{R}^W$ define low-rank strike and maturity factor matrices

$$A = \text{Linear}_{W \rightarrow N_K r}(h) \in \mathbb{R}^{N_K \times r}, \quad B = \text{Linear}_{W \rightarrow N_T r}(h) \in \mathbb{R}^{N_T \times r}.$$

The IV surface contribution from the grid head is

$$S_{ij}^{\text{grid}} = \sum_{\ell=1}^r A_{i\ell} B_{j\ell} + \beta_i^{(k)} + \beta_j^{(t)} + \beta^{(0)},$$

i.e. a rank- r bilinear form plus separable biases (we use $r = 24$). A parallel *residual head* — LayerNorm $\rightarrow$ Linear $\rightarrow$ SwiGLU $\rightarrow$ Dropout $\rightarrow$ Linear($N_K N_T$) — adds a dense correction; its dropout is safe because it feeds *directly* into the final linear with no downstream normalisation.

The output is passed through $0.01 + \text{softplus}_{\beta=3}$ so that predicted IVs are strictly bounded away from zero.

5.3 Pricable-region buffer

The tensor $\mathbf{1}_{\mathcal{P}} \in \{0, 1\}^{N_K \times N_T}$ (Figure 2) is registered as a persistent buffer of the model. At validation time the pricable IV-RMSE is

$$\text{RMSE}_{\mathcal{P}} = \sqrt{\frac{\sum_{ij \in \mathcal{P}} (\hat{\sigma}_{ij} - \sigma_{ij})^2}{|\mathcal{P}|}}.$$

6 Training

6.1 Vega-weighted IV-RMSE loss

Minimising a uniform IV-MSE over the grid over-weights the wings, where IV is numerically larger but economically less informative; a unit move in a deep-OTM IV corresponds to a sub-basis-point move in dollars. We therefore weight the loss by the Black–Scholes vega (at unit spot, and at the *target* IV rather than the prediction, so the weight is non-gradient):

$$w_{ij} = \sqrt{T_j} \phi(d_{1,ij}) e^{-qT_j}, \quad d_{1,ij} = \frac{-k_i + (r - q + \frac{1}{2}\sigma_{ij}^2)T_j}{\sigma_{ij}\sqrt{T_j}},$$

with ϕ the standard Gaussian density. Weights are normalised to have mean ≈ 1 over the pricable region. The per-surface data loss is

$$L_{\text{vega}}(\hat{\sigma}, \sigma, w, m, c) = \frac{\sum_{ij} w_{ij} m_{ij} c_{ij} (\hat{\sigma}_{ij} - \sigma_{ij})^2}{1 + \sum_{ij} w_{ij} m_{ij} c_{ij}}. \quad (2)$$

Here $m_{ij} \in \{0, 1\}$ is the quality mask ($\text{qm} \leq 1$) and $c_{ij} \in \{0.3, 1.0\}$ a confidence multiplier which down-weights cells that were filled by near-neighbour interpolation rather than priced directly. The +1 in the denominator is a small- N guard and has negligible effect once a reasonable number of cells are valid.

6.2 Soft arbitrage constraints

We add three unweighted-at-warm-start soft penalties, borrowed from the PINN literature [18], that regularise the surface toward an arbitrage-consistent shape:

Calendar arbitrage. Total implied variance $\text{TV}(k, T) = \sigma^2(k, T) \cdot T$ must be non-decreasing in T at each fixed k :

$$L_{\text{cal}} = \mathbb{E}[\text{ReLU}(\text{TV}_{T_j} - \text{TV}_{T_{j+1}})^2] + \text{long-range terms (1W} \rightarrow \text{1M, 1M} \rightarrow \text{6M, 6M} \rightarrow \text{2Y)}.$$

Durrleman butterfly. Let $w(k, T) = \sigma^2(k, T)T$ and define the Durrleman function

$$g(k, T) = \left(1 - \frac{k \partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{1}{2} \partial_k^2 w.$$

Butterfly no-arbitrage is equivalent to $g \geq 0$ [5]. We penalise $L_{\text{bfly}} = \mathbb{E}[\text{ReLU}(-g)^2]$ over interior strikes.

Term-structure slope. The difference of ATM IVs between the shortest and longest traded maturities is a convenient calibration statistic that κ drives almost entirely. We add an L2 penalty on its deviation from the target: $L_{\text{ts}} = \mathbb{E}[(\sigma_{\text{ATM, long}}^{\text{NN}} - \sigma_{\text{ATM, short}}^{\text{NN}}) - (\sigma_{\text{ATM, long}}^* - \sigma_{\text{ATM, short}}^*)^2]$.

Total loss. With $\lambda_{\text{cal}} = 10^{-2}$, $\lambda_{\text{bfly}} = 5 \times 10^{-3}$, $\lambda_{\text{ts}} = 3 \times 10^{-3}$ (ramped linearly over epochs 15–35 to avoid drowning the data term early on), we minimise

$$L = L_{\text{vega}} + \lambda_{\text{cal}} L_{\text{cal}} + \lambda_{\text{bfly}} L_{\text{bfly}} + \lambda_{\text{ts}} L_{\text{ts}}.$$

6.3 Optimiser, schedule and mixed precision

Adam [15] with $(\beta_1, \beta_2) = (0.9, 0.999)$, learning rate 3×10^{-4} , weight decay 10^{-5} , and `ReduceLROnPlateau` (factor 0.5, patience 10). Gradients are clipped to unit norm. Forward passes run in BF16 via PyTorch autocast while losses are reduced in FP32 for under-flow safety. Validation runs end-to-end

in FP32. Early stopping monitors the total validation loss with patience 30 starting at epoch 25, by which time the PINN warmup has fully ramped.

Training uses batch size 8192, with the entire dataset preloaded into host RAM (`-preload all`) to remove an early IO bottleneck; with this, one epoch takes ≈ 15.00 s on a single GPU and the full 100.00-epoch run completes in ≈ 25.00 min.

6.4 A non-obvious bug: LayerNorm after Dropout

During the first baseline run the training loss fell to 62.00 bps while the validation loss stalled at 199.00 bps — a $3.2\times$ gap not explicable by dataset drift (both HDF5 files were sampled from the identical prior) nor by overfitting (the gap appeared from the first epoch and never closed). Loading the best checkpoint and evaluating in `.eval()` mode on a 20 000.00-row subset of the *training* file reproduced the 199.00 bps number exactly, identifying the gap as *mode-dependent* rather than data-dependent.

Root cause: the stem contained a `nn.Dropout` [16] layer that fed directly into the first ResBlock, whose first operation is `self.ln(x)` [17] (Section 5.1). PyTorch’s inverted dropout multiplies surviving activations by $1/(1-p)$ at training time so that linear layers see activations of the same *mean and variance* in train and eval modes. LayerNorm does not see the same statistics because it recomputes mean/variance *per sample*; a feature whose pre-LN value is c becomes $c/(1-p)$ with probability $1-p$ and 0 with probability p , and the LN normalisation of this bimodal mixture does not agree with the LN normalisation of the original constant c . The effect compounds across six ResBlocks.

Fix: remove stem dropout entirely. For symmetry, the shared- z dropout inside `GridFactorizedHead` was also removed, since the bilinear einsum $A \cdot B$ mixes two asymmetric projections of the same zeroed-then-scaled input. The residual head’s dropout was *kept*: it sits immediately before a final `Linear` with no downstream normalisation, so the invariance argument is recovered. After the fix, training and validation stayed aligned to within 1.50 bps throughout the run.

7 Results

7.1 Training convergence

Figure 3 shows IV-RMSE on the train set (derived from the vega loss, which coincides with IV-RMSE up to the vega weighting) and on the pricable region of the validation set, along with the three soft-constraint penalties on log scale. The validation loss bottoms out at 35.99 bps at epoch 96 (35.73 bps on the pricable region only, coverage 96.7%).

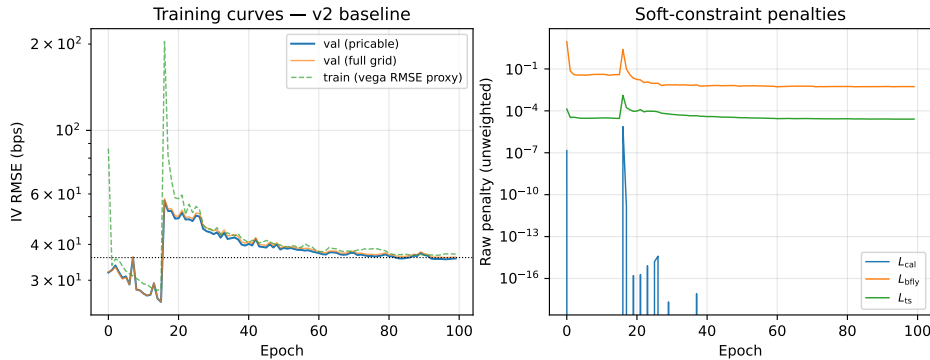


Figure 3: Training curves for the v2 baseline. Left: IV-RMSE (log scale) for train (vega proxy, dashed), validation on the full 49×14 grid, and validation restricted to the pricable region $\mathcal{P}$. Dotted horizontal line at 35.99 bps marks the best validation value. Right: unweighted soft-constraint penalties; all three decay smoothly under the ramp schedule with no evidence of them dominating the data term.

7.2 Surface examples

Figure 4 overlays the COS pricer, Andersen QE MC, and surrogate on a representative parameter set (the one with the highest pricable coverage among the 200 MC sets). The surrogate reproduces the skew and calendar structure cleanly; residuals concentrate in the deep-OTM short- T corner, exactly where both the COS window and the MC scheme are themselves least reliable.

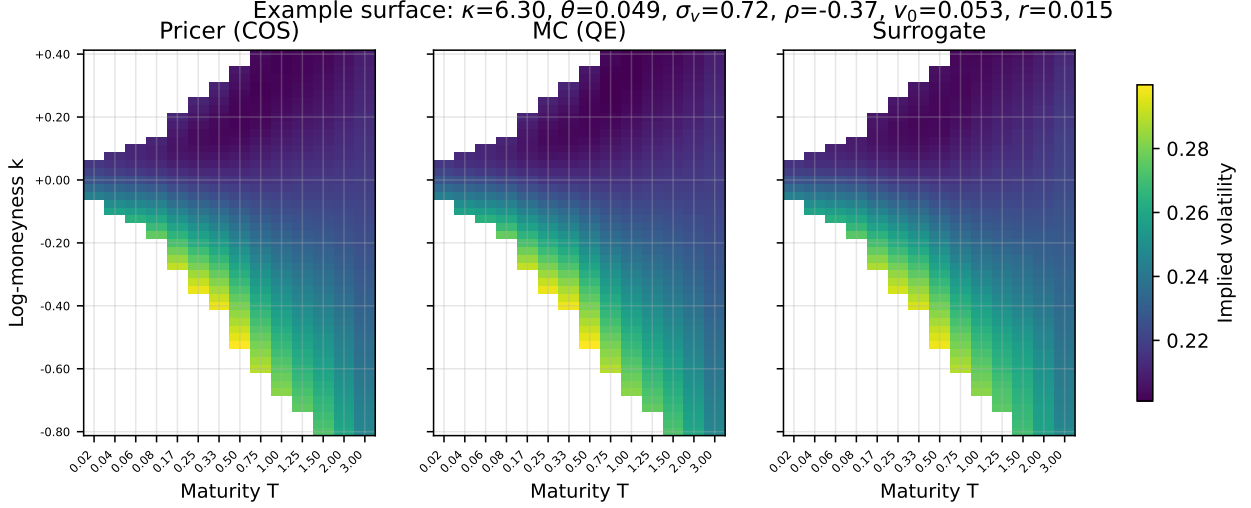


Figure 4: A representative IV surface: pricer (left), MC reference (centre), and surrogate (right). The three surfaces share the same colour scale; non-pricable cells are blanked. The parameters are shown in the figure title.

7.3 Gate vs MC reference

Figure 5 summarises the IV-difference distributions across all 84 747.00 pricable-and-finite cells of the 200 MC surfaces.

Comparison	IV-RMSE (bps)	$p_{95} \Delta\sigma $	$p_{99} \Delta\sigma $
Model vs MC	36.67	73	143
Model vs pricer	36.49	72	142
MC vs pricer	3.44	6.1	11.3

Table 1: Pairwise IV statistics on the 200-surface MC reference.

Two observations: (i) *Model vs pricer* is indistinguishable from *model vs MC* because the MC noise floor (~ 3.00 bps) is an order of magnitude below the surrogate residual (~ 36.00 bps). (ii) The 50.00 bps gate holds on 89.58% of cells, below the 95% target. The miss is entirely in the p_{99} tail (143.00 bps); closing it would require either a deeper surrogate, more training data at the tails, or sharper short- T numerics. This is future work rather than a bug.

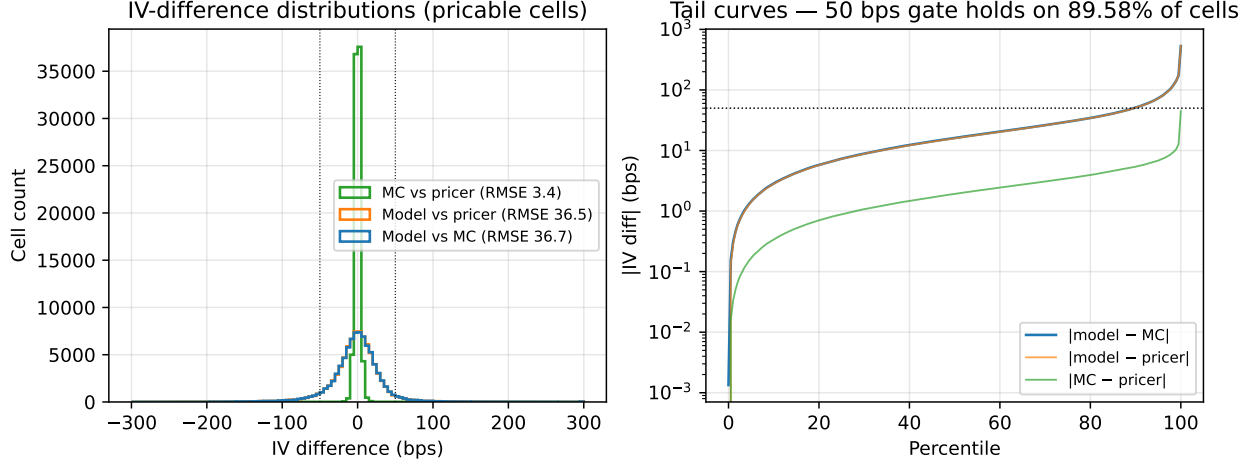


Figure 5: Left: IV-difference histograms on pricable cells. The MC-vs-pricer distribution (green) defines the noise floor. The model-vs-MC and model-vs-pricer distributions are virtually identical, confirming that surrogate error dominates MC error. Dotted verticals at ± 50.00 bps mark the gate. Right: quantile curves of the absolute differences ($\log y$). The horizontal line at 50.00 bps is crossed at the 89.6th percentile for $|\text{model} - \text{MC}|$.

7.4 Calibration sanity

We check that the surrogate is usable for calibration by running a gradient-descent inversion on the 200 MC surfaces. We parametrise the search in the latent $z \in \mathbb{R}^6$ with $\theta = \theta_{\min} + (\theta_{\max} - \theta_{\min})\sigma(z)$ (element-wise sigmoid), initialise every z_k at 0 (i.e. the centre of each parameter’s prior range), and run Adam for 600 iterations at learning rate 5×10^{-2} on the pricable MSE between $\Phi_\phi(\theta)$ and the observed MC IV. All 200 surfaces are calibrated in a single batched run in a few seconds on GPU. Table 2 and Figure 6 report the outcome. Two kinds of parameters emerge:

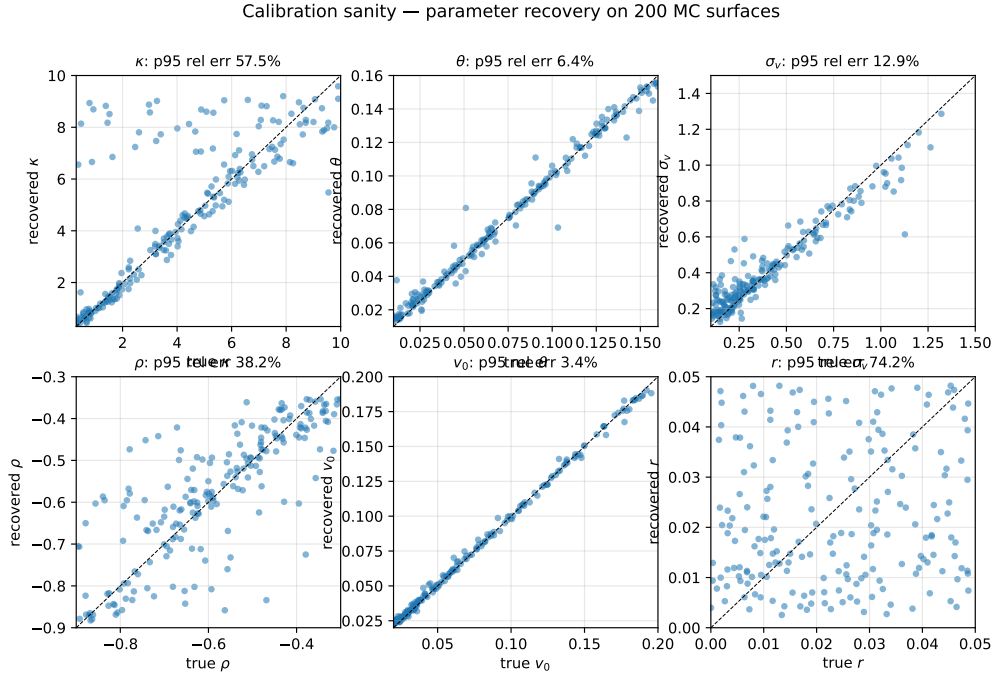


Figure 6: Recovered versus true parameters on the 200-surface MC reference after a flat-init Adam inversion of the surrogate. Dashed lines are the identity. Per-parameter p_{95} relative errors are reported in the titles.

Parameter	Range	Mean $ \Delta $	$p_{95} \Delta $	p_{95} rel
κ	[0.30, 10.0]	1.014	5.58	57.5%
θ	[0.01, 0.16]	0.003	0.010	6.4%
σ_v	[0.10, 1.50]	0.060	0.180	12.9%
ρ	[-0.90, -0.30]	0.068	0.229	38.2%
v_0	[0.02, 0.20]	0.002	0.006	3.4%
r	[0.00, 0.05]	0.015	0.037	74.2%

Table 2: Parameter recovery error. p_{95} relative is $p_{95}|\Delta|$ divided by the range width.

The *well-identified* parameters v_0, θ, σ_v recover with p_{95} relative errors of 3.4%, 6.4%, 12.9%; these are the quantities that actually dictate the short- and long-maturity ATM levels and the convexity of the skew. The *poorly-identified* set κ, ρ, r reflects the classical Heston identifiability problem: the pair (κ, ρ) trades off along the leverage direction (a higher mean-reversion can be compensated by a more negative correlation) and the rate r is nearly degenerate with v_0 in the ATM region of a single-curve, flat-rate calibration. In production calibration, r would be fixed from the yield curve and κ regularised or constrained via a prior.

The post-fit per-surface IV-RMSE averages 30.74 bps (median 25.82 bps, $p_{95} = 70.11$ bps) — below the surrogate’s own flat-prior validation error on most surfaces, confirming that the surrogate is smooth and invertible under gradient descent.

8 Discussion and future work

8.1 What this project establishes

The pipeline produces a fully differentiable Heston IV surrogate whose end-to-end error is dominated by a single-checkpoint training run of ~ 25 minutes on a single GPU; the MC gate evaluates consistency against an independent SDE solver and the calibration sanity check confirms the model is usable as a calibration target.

It is thus now used as the calibration engine for the Continuous-Time Markov Chain (CTMC) pricer for american call and puts under Heston SLV, which is available in my options pricer project in Rust <https://github.com/Mttmin/options-pricer>.

8.2 What it does not yet achieve

The 50.00 bps gate misses the 95% target by 5.4 percentage points. Plausible remedies, ordered by expected return on effort:

1. Deeper trunk ($N_B = 10\text{--}12$) and/or wider head ($r = 64$), with a proportionally longer training schedule.
2. Focal re-weighting of the loss at the high- p_{99} tails (short maturities, deep wings) once per-cell residuals are tracked.
3. Independent loss on the pricable-region residual head only, to stop the grid head drifting on unpricable cells that leak into gradients during training.

Concerning the calibration side, the three poorly-identified parameters κ, ρ, r are not a surrogate problem and would be handled at the calibration layer by:

1. fixing r from the discount curve;
2. imposing an informative prior on κ (e.g. log-normal centred at the historical mean-reversion);

3. multi-start Adam from ~ 5 – 10 seeds and keeping the lowest-RMSE minimum, which would dampen the κ - ρ tradeoff for well-conditioned surfaces.

Furthermore, while this architecture is used here for a Heston stochastic local volatility surrogate, the same pipeline could be applied to more complex models (e.g. Bates jump-diffusion, or Dense SLV) and to multi-asset surfaces, where the grid-factorised head would be even more beneficial.

8.3 Reproducibility

All artefacts are checked into the repository:

Artefact	Path
Trained checkpoint	runs/v2_baseline/best.pt
Training log	runs/v2_baseline/train_log.csv
Training HDF5	data/heston_pipeline_v2.h5
Validation HDF5	data/heston_pipeline_v2_val.h5
MC reference	data/heston_mc_reference.h5
MC-gate diagnostic	.../diagnostics/evaluate_v2_vs_mc.py
Calibration sanity	.../diagnostics/calibrate_sanity.py
Figure script	report/make_figures.py
Results summary	SCENARIO_C_RESULTS.md

References

- [1] S. L. Heston, *A closed-form solution for options with stochastic volatility with applications to bond and currency options*, Review of Financial Studies 6 (1993), 327–343.
- [2] H. Albrecher, P. Mayer, W. Schoutens, J. Tistaert, *The little Heston trap*, Wilmott Magazine, January 2007, 83–92.
- [3] F. Fang, C. W. Oosterlee, *A novel pricing method for European options based on Fourier-cosine series expansions*, SIAM Journal on Scientific Computing 31 (2008), 826–848. doi:[10.1137/080718061](https://doi.org/10.1137/080718061).
- [4] L. Andersen, *Simple and efficient simulation of the Heston stochastic volatility model*, Journal of Computational Finance 11 (2008), 1–42.
- [5] V. Durrleman, *From implied to spot volatilities*, PhD thesis, Princeton University (2004); see also Finance and Stochastics 14 (2010) 157–177.
- [6] P. Carr, D. Madan, *Option valuation using the fast Fourier transform*, Journal of Computational Finance 2 (1999), 61–73.
- [7] A. Hernandez, *Model calibration with neural networks*, Risk Magazine, September 2017. SSRN: [2812140](https://ssrn.com/abstract=2812140).
- [8] B. Horvath, A. Muguruza, M. Tomas, *Deep learning volatility: a deep neural network perspective on pricing and calibration in (rough) volatility models*, Quantitative Finance 21 (2021), 11–27. doi:[10.1080/14697688.2020.1817974](https://doi.org/10.1080/14697688.2020.1817974). arXiv:[1901.09647](https://arxiv.org/abs/1901.09647).
- [9] C. Bayer, B. Stemper, *Deep calibration of rough stochastic volatility models*, arXiv:[1810.03399](https://arxiv.org/abs/1810.03399), 2018.
- [10] S. Liu, A. Borovikh, L. Grzelak, C. W. Oosterlee, *A neural network-based framework for financial model calibration*, Journal of Mathematics in Industry 9 (2019), 9. doi:[10.1186/s13362-019-0066-7](https://doi.org/10.1186/s13362-019-0066-7). arXiv:[1904.10523](https://arxiv.org/abs/1904.10523).
- [11] J. Ruf, W. Wang, *Neural networks for option pricing and hedging: a literature review*, Journal of Computational Finance 24 (2020), 1–46. arXiv:[1911.05620](https://arxiv.org/abs/1911.05620).
- [12] G. Dimitroff, D. Röder, C. P. Fries, *Volatility model calibration with convolutional neural networks*, SSRN: [3252432](https://ssrn.com/abstract=3252432), 2018.
- [13] F. Baschetti, G. Bormetti, S. Rossi, *Deep calibration with random grids*, Quantitative Finance 24 (2024), 9. doi:[10.1080/14697688.2024.2332375](https://doi.org/10.1080/14697688.2024.2332375).
- [14] P. Büchel, M. Kratochwil, N. Nagl, D. Rösch, *Deep calibration of financial models: turning theory into practice*, Review of Derivatives Research 25 (2022), 109–136.
- [15] D. P. Kingma, J. Ba, *Adam: a method for stochastic optimization*, ICLR 2015. arXiv:[1412.6980](https://arxiv.org/abs/1412.6980).
- [16] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, *Dropout: a simple way to prevent neural networks from overfitting*, Journal of Machine Learning Research 15 (2014), 1929–1958.
- [17] J. L. Ba, J. R. Kiros, G. E. Hinton, *Layer normalization*, arXiv:[1607.06450](https://arxiv.org/abs/1607.06450), 2016.
- [18] M. Raissi, P. Perdikaris, G. E. Karniadakis, *Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs*, Journal of Computational Physics 378 (2019), 686–707. doi:[10.1016/j.jcp.2018.10.045](https://doi.org/10.1016/j.jcp.2018.10.045).

- [19] J. Gatheral, *A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives*, Presentation at Global Derivatives & Risk Management, Madrid, 2004.